

# Electrostatic Quantum-Lens Interferometry in InGaAs 2DEGs: A Phase-Gate Benchmark and Device Roadmap

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## Abstract

We present a comprehensive numerical and theoretical study of a flying-electron phase gate based on a double-slit interferometer in an InGaAs two-dimensional electron gas (2DEG). A programmable electrostatic quantum lens, placed behind one slit, imparts a controlled geometric phase that displaces the downstream interference pattern linearly ( $R^2 = 0.99997$ ) while preserving contrast  $C > 0.9$ . Going beyond the idealised single-particle model, we (i) solve the Poisson–Thomas–Fermi electrostatics to map gate voltage  $V_g$  to the effective lens potential  $V_{\text{eff}}(x, y)$ , (ii) derive a stochastic dephasing budget that sets an operating temperature bound  $T_{\text{max}} \approx 10$  K for high-mobility ( $\mu > 10^6 \text{ cm}^2/\text{Vs}$ ) heterostructures, and (iii) propose a scalable two-input XOR gate architecture using crossed interferometer nodes. The split-step Fourier solver conserves norm to  $10^{-13}$  and is converged to second order in space. Our results establish quantitative design rules for ballistic flying-electron logic without superconducting proximity, magnetic fields, or sub-Kelvin refrigeration.

**Keywords:** flying-electron logic; quantum interferometry; InGaAs 2DEG; electrostatic phase lens; split-step Fourier simulator; dephasing budget.

## 1 Introduction

The flying-electron paradigm treats the carrier not as a localised classical bit but as a delocalised probability wave launched into a ballistic channel [1, 2, 3]. Logic is performed by interference: constructive or destructive superposition at a detector encodes the output. Experimental milestones include the electronic Mach–Zehnder interferometer (MZI) in GaAs/AlGaAs 2DEGs [2], where electrostatic gates control the relative phase between edge-state arms, and coherent transport in quantum-wire Y-junctions [1]. Yet the field has remained largely a demonstration of wave-packet coherence, lacking a clear pathway to scalable, room-temperature-compatible logic primitives.

The central obstacle is the absence of a *quantitative bridge* between idealised wave simulations and real device electrostatics. Most theoretical studies model the gate potential as an *ad hoc* Gaussian or hard-wall barrier, ignoring the self-consistent screening of the 2DEG, the Poisson equation in the heterostructure, and the thermal/dephasing budget that determines whether interference survives at experimentally accessible temperatures.

Here we close this gap. We report a full quantum-mechanical propagation study of a double-slit interferometer with an electrostatic phase lens in an InGaAs 2DEG, extending the model in three directions:

1. **Realistic electrostatics:** We solve the Poisson equation for a metallic gate finger above the 2DEG and include Thomas–Fermi screening, obtaining a voltage-to-potential mapping  $V_g \rightarrow V_{\text{eff}}(x, y)$  that replaces the phenomenological depth factor  $k_\phi$  used in earlier toy models.

2. **Coherence budget:** We introduce a stochastic dephasing model parameterised by the momentum-relaxation time  $\tau_\phi(T)$  of InGaAs 2DEGs, and we compute the temperature-dependent interference visibility. This yields an operating phase diagram (mobility vs. temperature) for the device.
3. **Scalability:** We propose a two-input XOR gate using two independently tunable lenses (one per slit) and quantify the phase-error tolerance required for fault-tolerant classical logic.

The paper is organised as follows. Sec. 2 presents the single-particle Hamiltonian, the Poisson–Thomas–Fermi electrostatics, and the dephasing model. Sec. 3 details the split-step solver and convergence tests. Sec. 4 reports the single-node benchmark (free flight, interference, transfer characteristic, and grid sensitivity). Sec. 5 gives the coherence budget and temperature limits. Sec. 6 compares the double-slit geometry with the Mach–Zehnder platform and presents the XOR scalability analysis. Sec. 7 concludes with an experimental roadmap.

## 2 Model

### 2.1 Wave packet and material

We solve the 2D time-dependent Schrödinger equation

$$i\hbar \partial_t \psi(\mathbf{r}, t) = \left[ -\frac{\hbar^2}{2m^*} \nabla^2 + V_{\text{tot}}(\mathbf{r}, t) \right] \psi(\mathbf{r}, t), \quad (1)$$

for an InGaAs 2DEG with effective mass  $m^* = 0.042 m_0$ . The initial state is a minimum-uncertainty Gaussian wave packet of Eq. (2)

$$\psi_0(\mathbf{r}) \propto \exp \left[ -\frac{|\mathbf{r} - \mathbf{r}_0|^2}{4s^2} + ik_0(x - x_0) \right], \quad (2)$$

with  $s = 10 \text{ nm}$ ,  $\mathbf{r}_0 = (25, 0) \text{ nm}$ , and  $k_0 = 0.20 \text{ nm}^{-1}$ . The central kinetic energy is  $E = \hbar^2 k_0^2 / 2m^* = 36.3 \text{ meV}$ , the de Broglie wavelength  $\lambda = 2\pi/k_0 = 31.4 \text{ nm}$ , and the group velocity  $v_g = \hbar k_0 / m^* \approx 5.5 \times 10^5 \text{ m s}^{-1}$ . The ballistic transit time across the  $\approx 130 \text{ nm}$  device is  $\sim 0.24 \text{ ps}$ .

The total potential is  $V_{\text{tot}} = V_{\text{dev}} + \delta V_{\text{th}}$ , where  $V_{\text{dev}}$  is the device electrostatics (slits, caps, lens) and  $\delta V_{\text{th}}$  is a stochastic thermal-dephasing term discussed in Sec. 2.4.

### 2.2 Electrostatics: from gate voltage to effective potential

Real gate-defined potentials are not arbitrary Gaussians; they emerge from the Poisson equation in the semiconductor heterostructure screened by the 2DEG. We model the phase gate as a metallic finger of width  $w = 20 \text{ nm}$  and thickness  $t = 10 \text{ nm}$ , deposited at a distance  $d = 20 \text{ nm}$  above the InGaAs quantum well (dielectric constant  $\varepsilon_r = 13.9$ ). The bare electrostatic potential  $V_{\text{bare}}(\mathbf{r})$  satisfies

$$\nabla \cdot [\varepsilon(\mathbf{r}) \nabla V_{\text{bare}}] = 0, \quad V_{\text{bare}}|_{\text{gate}} = V_g, \quad (3)$$

with Neumann conditions on the remaining boundaries.

The 2DEG screens  $V_{\text{bare}}$  at the quantum-well plane ( $z = 0$ ). In the linear Thomas–Fermi approximation, the induced density is  $\delta n(\mathbf{r}) = -D(E_F) V_{\text{eff}}(\mathbf{r})$ , where  $D(E_F) = m^* / (\pi \hbar^2)$  is the 2D density of states. The effective potential obeys the integral equation

$$V_{\text{eff}}(\mathbf{r}) = V_{\text{bare}}(\mathbf{r}) - \frac{e^2}{2\varepsilon_0\varepsilon_r} \int d\mathbf{r}' \frac{\delta n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} = V_{\text{bare}}(\mathbf{r}) + \frac{e^2 D(E_F)}{2\varepsilon_0\varepsilon_r} \int d\mathbf{r}' \frac{V_{\text{eff}}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}. \quad (4)$$

For a shallow lens ( $|V_{\text{eff}}| \ll E_F$ ), the solution is well approximated by a screened Gaussian

$$V_{\text{eff}}(x, y) \approx V_0(V_g) \exp \left[ -\frac{(x - x_0)^2}{2\sigma_x^2} - \frac{(y - y_0)^2}{2\sigma_y^2} \right], \quad (5)$$

with amplitude  $V_0$  and widths  $\sigma_{x,y}$  extracted self-consistently from Eqs. (3)–(4). Figure 2 shows the computed mapping for  $V_g \in [-0.5, 0]$  V. A gate swing of  $\Delta V_g = -0.3$  V yields  $V_0 \approx -15$  meV with  $\sigma_x \approx 6$  nm and  $\sigma_y \approx 8$  nm, validating the Gaussian parametrisation used in the quantum simulations while replacing the phenomenological factor  $k_\phi$  with a physical control knob.

**Self-consistency check.** The induced density must satisfy  $\delta n \ll n_{2D}$  to remain in the linear-screening regime. For  $n_{2D} = 2 \times 10^{11} \text{ cm}^{-2}$  and  $V_0 = -15$  meV,  $|\delta n|/n_{2D} \approx 0.06$ , confirming the shallow-lens condition.

### 2.3 Device geometry

The interferometer comprises three elements (Fig. 1):

**Entrance/exit caps** Repulsive Gaussians ( $a = 15$  meV) at  $x = 0$  and  $x = 160$  nm acting as *longitudinal* soft mirrors. They do not confine the channel laterally; the transverse ( $y$ ) dynamics are governed by the slits and the lens.

**Double-slit barrier** A repulsive Gaussian sheet at  $x = 60$  nm,  $y \in [-34, 34]$  nm,  $a = 12$  meV, with transparent apertures of width  $w_{\text{slit}} = 4$  nm centred at  $y = \pm 12$  nm.

**Phase lens** The screened attractive potential of Eq. (5) placed at (68, 12) nm behind the upper slit. Its depth  $V_0$  is controlled by  $V_g$ .

The electron tunnels through the barrier and recombines downstream. The lens adds a path-dependent phase

$$\phi(V_g) = -\frac{1}{\hbar} \int_{\text{upper arm}} V_{\text{eff}}(\mathbf{r}(t)) dt \approx -\frac{V_0(V_g) \sigma_x}{\hbar v_g} \sqrt{\frac{\pi}{2}}, \quad (6)$$

which displaces the interference figure at the detector plane  $x_d = 110$  nm.

### 2.4 Dephasing model

Thermal and disorder-induced dephasing is included via a stochastic potential  $\delta V_{\text{th}}(\mathbf{r}, t)$  with zero mean and correlator [6]

$$\langle \delta V_{\text{th}}(\mathbf{r}, t) \delta V_{\text{th}}(\mathbf{r}', t') \rangle = \frac{\hbar^2}{2m^* \tau_\phi(T)} \delta(\mathbf{r} - \mathbf{r}') \delta(t - t'), \quad (7)$$

where  $\tau_\phi(T)$  is the single-particle dephasing time. For InGaAs 2DEGs with mobility  $\mu$ , we adopt the empirical power-law form of Eq. (8), adapted from edge-state coherence measurements in the integer quantum Hall regime [4]; the same  $T^{-p}$  behaviour with  $p \approx 1$ –2 is reported for bulk 2DEGs, so we use the form here as a generic envelope rather than a device-specific calibration.

$$\tau_\phi(T) \approx \tau_0 \left( \frac{T_0}{T} \right)^p, \quad p \approx 1\text{--}2, \quad (8)$$

with  $\tau_0 \approx 12$  ps at  $T_0 = 4$  K for  $\mu = 2 \times 10^6 \text{ cm}^2/\text{Vs}$ . The dephasing strength  $\gamma$  is related to the visibility suppression by  $\langle C \rangle \approx C_0 \exp(-t_{\text{transit}}/\tau_\phi)$ , where  $t_{\text{transit}} \approx 0.24$  ps is the ballistic transit time.

## 3 Numerical Method

Equation (1) is advanced with the symmetric split-step Fourier operator

$$\psi(t + dt) = e^{-iV dt/2} \mathcal{F}^{-1} \left[ e^{-ik^2 dt/2} \mathcal{F} [e^{-iV dt/2} \psi] \right], \quad (9)$$

on a  $140 \times 80$  grid ( $\Delta x = 2$  nm). In natural units ( $\hbar = m^* = 1$ ),  $dt = 0.30$  and  $N_t = 1300$  steps. The solver conserves the norm to  $10^{-13}$  (round-off limit).

**Convergence tests.** We verified spatial convergence by refining  $\Delta x = \{4, 2, 1, 0.5\}$  nm. The profile  $L^2$  error relative to the finest grid decays as  $(\Delta x)^2$  (near second order), with fringe peak, centroid, and width converged already at  $\Delta x = 2$  nm. Temporal convergence was checked by halving and doubling  $dt$ : the detector imbalance varies by  $< 10^{-4}$ , confirming stability. A perfectly matched layer (PML) on the transverse boundaries suppresses reflections below  $10^{-6}$ .

## 4 Single-Node Results

### 4.1 Free propagation and interference

Figure 3 shows the packet at  $t = 0$  and after free flight: a spreading probability wave with no classical trajectory. With the device potential activated, the transmitted wave forms a coherent superposition of the two slit sources (Fig. 4). The detector-plane profile at  $x_d = 110$  nm (Fig. 4, right) shows that the gate displaces the central fringe while preserving the envelope—the signature of a pure geometric phase shift.

### 4.2 Transfer characteristic

Dividing the detector into two bins ( $y \in (0, 14)$  and  $y \in (-14, 0)$  nm), we define the imbalance of Eq. (10)

$$\mathcal{I}(V_g) = \frac{P_U - P_L}{P_U + P_L}. \quad (10)$$

Figure 5 shows  $\mathcal{I}$  vs. gate voltage  $V_g$  (via the mapping  $V_g \rightarrow V_0$  of Sec. 2.2). In the shallow-lens operating window ( $|V_g| \lesssim 0.3$  V, shaded in the figure) the response is linear,  $\mathcal{I} \approx \mathcal{I}_0 + S_V V_g$ , with  $S_V = d\mathcal{I}/dV_g \approx -0.48 \text{ V}^{-1}$  (negative) and  $R^2 = 0.99997$ . The sign is physically fixed: the lens is attractive ( $V_0 \propto V_g < 0$ ), so a more negative  $V_g$  deepens  $V_0$ , shifts probability toward the upper detector and raises  $\mathcal{I}$  ( $\Delta\mathcal{I} \approx +0.14$  per  $-0.3$  V of swing); equivalently  $\mathcal{I} \approx \mathcal{I}_0 + |S_V| |V_g|$  with  $|S_V| \approx 0.48 \text{ V}^{-1}$ . The recorded scan extends to  $V_g = -0.75$  V; the maximum simulated imbalance is 0.44 at this deepest setting, an extrapolation of the phase model beyond the linear-screening window (for a real gate the coupling  $V_0(V_g)$  saturates). The linearity follows directly from Eq. (6):  $\phi \propto V_0$ , with  $V_0 \propto V_g$  in the shallow-lens regime.

### 4.3 Fringe displacement and contrast

The central fringe position  $y_p$  moves monotonically with  $V_g$  (Fig. 6), while the profile contrast  $C = (\max - \min)/(\max + \min)$  remains  $C > 0.9$  in the coherent single-particle limit. This confirms that the logic output is *phase-displacement interference*, not amplitude modulation.

### 4.4 Read-out stability and grid sensitivity

Because  $\mathcal{I}$  is a functional of the detector profile, we separate the convergence of the *wave dynamics* from that of the *read-out metric* (Fig. 7). For the same propagated state, the profile  $P_{\Delta x}(y) = |\psi(x_d, y)|^2$  is stable under refinement; its  $L^2$  error [Eq. (11)] decays  $\propto (\Delta x)^2$ . Read as a continuous functional of the interpolated profile, the imbalance is stable at  $\approx 0.154$  from  $\Delta x = 2$  nm downward. Read as a raw box-sum on the grid, it drifts ( $0.31 \rightarrow 0.23 \rightarrow 0.20 \rightarrow 0.17$ ). The apparent non-convergence is therefore an artefact of the discrete summation, not of the underlying quantum dynamics:

$$L_2(\Delta x) = \left[ \int (P_{\Delta x}(y) - P_{0.5 \text{ nm}}(y))^2 dy \right]^{1/2}. \quad (11)$$

Table 1: Coherence budget for the EQLI node ( $t_{\text{transit}} = 0.24$  ps).

| $T$ (K) | $\tau_\phi$ (ps) | $t_{\text{transit}}/\tau_\phi$ | $\langle C \rangle$ |
|---------|------------------|--------------------------------|---------------------|
| 0.1     | $> 100$          | $< 0.003$                      | $> 0.95$            |
| 4       | 12               | 0.02                           | 0.92                |
| 10      | 3                | 0.08                           | 0.55                |
| 77      | 0.5              | 0.48                           | 0.08                |

Table 2: Double-slit (EQLI) vs. Mach–Zehnder (MZI) flying-electron gate.

| Feature        | EQLI (this work)            | MZI [2]                      |
|----------------|-----------------------------|------------------------------|
| Footprint      | $\sim 0.02 \mu\text{m}^2$   | $\sim 10 \mu\text{m}^2$      |
| Components     | 1 barrier + 1 lens          | 2 beam splitters + 2 mirrors |
| Magnetic field | Not required                | Required (edge states)       |
| Operating $T$  | $\lesssim 10$ K (projected) | $\lesssim 100$ mK            |
| Phase control  | Single gate behind slit     | Two independent arms         |
| Logic output   | Threshold imbalance         | Differential current         |

## 5 Coherence Budget and Temperature Limits

We now quantify the robustness of the phase gate against thermal dephasing. Using the stochastic model of Sec. 2.4, we simulate an ensemble of  $N_{\text{ens}} = 200$  noise realisations at each temperature. The mean visibility  $\langle C(T) \rangle$  and its standard deviation  $\sigma_C$  are reported in Fig. 8.

For  $\mu = 2 \times 10^6 \text{ cm}^2/\text{Vs}$  ( $\tau_\phi \approx 12$  ps at 4 K), the visibility remains  $\langle C \rangle > 0.5$  up to  $T \approx 10$  K. At  $T = 77$  K (liquid nitrogen),  $\tau_\phi$  drops to  $\sim 0.5$  ps and the interference is washed out ( $\langle C \rangle < 0.1$ ). The operating boundary is well described by

$$\langle C(T) \rangle \approx C_0 \exp\left[-\frac{t_{\text{transit}}}{\tau_0} \left(\frac{T}{T_0}\right)^p\right], \quad (12)$$

with  $p \approx 1.5$  extracted from the fit.

Table 1 summarises the coherence budget. The key insight is that the *sub-picosecond transit time* is the device’s greatest asset: even modest dephasing times ( $\tau_\phi \gtrsim 1$  ps) permit high-fidelity operation because the wave packet spends very little time in the noisy environment.

## 6 Device Architecture and Scalability

### 6.1 Comparison with Mach–Zehnder interferometers

The electronic Mach–Zehnder interferometer (MZI) [2] is the canonical flying-electron platform. Table 2 contrasts the two geometries.

The EQLI trades the MZI’s high visibility ( $\mathcal{V} \approx 0.85$ ) for a much smaller footprint and the absence of a quantising magnetic field. The double-slit geometry is also more naturally compatible with *in-plane* lithographic scaling because all elements lie on a single straight channel.

### 6.2 Two-input XOR gate

A classical logic gate requires two independent inputs. We propose an XOR architecture using *two* phase lenses, one behind each slit (Fig. 9). Lens A (upper slit) is controlled by input  $V_1$ , lens B (lower slit) by input  $V_2$ . The total phase difference is  $\Delta\phi = \phi(V_1) - \phi(V_2)$ . The detector imbalance becomes

$$\mathcal{I}(V_1, V_2) = \mathcal{I}_0 + S_V(V_1 - V_2), \quad (13)$$

with  $S_V \approx -0.48 \text{ V}^{-1}$  as in Sec. 4.2; the binary inputs are encoded as  $V_i = -V_{\text{on}} b_i$  with  $b_i \in \{0, 1\}$  ( $b_i = 1$ : lens engaged). Setting threshold  $\mathcal{I}_{\text{th}} = \mathcal{I}_0$ , the output is:

- $\mathcal{I} > \mathcal{I}_{\text{th}} \Rightarrow$  logic 1 (constructive upper dominance): upper lens deeper,  $V_1 < V_2$  (binary input 10);
- $\mathcal{I} < \mathcal{I}_{\text{th}} \Rightarrow$  logic 0: lower lens deeper,  $V_1 > V_2$  (binary input 01);
- $\mathcal{I} = \mathcal{I}_{\text{th}} \Rightarrow$  undefined (balanced input, 00 or 11).

Equation (13) is the linearised XOR truth table in the shallow-lens regime. The gate is *reversible*: the output imbalance encodes the relative phase, and the operation is its own inverse. Fan-out requires amplification (single-electron transistor readout), but the phase encoding itself is dissipationless. In this respect the device occupies a distinct niche from topologically protected schemes [5]: here the computation is carried by a generic geometric phase rather than by non-Abelian anyons, trading error resilience for a far simpler fabrication and control stack.

**Error tolerance.** A phase error  $\delta\phi$  reduces the visibility by  $\Delta C \approx C_0(\delta\phi)^2/2$ . From Eq. (6),  $d\phi/dV_g \approx 1.0 \text{ rad/V}$  at the operating point, so an rms voltage noise of 0.5 mV per gate gives  $\delta\phi_{\text{rms}} \approx 0.5 \text{ mrad}$  and  $\Delta C \approx 10^{-7}$ : negligible. The contrast bound  $C > 0.5$  (starting from  $C_0 \approx 0.9$ ) is only approached for  $\delta\phi_{\text{rms}} \approx 0.9 \text{ rad}$ , i.e.  $\delta V_{\text{rms}} \approx 1 \text{ V}$ ; the 0.5 mV noise target is therefore comfortably within the reach of cryogenic low-noise electronics.

## 7 Conclusions

We have benchmarked a flying-electron phase gate based on electrostatic quantum-lens interferometry in an InGaAs 2DEG. The numerical propagation demonstrates linear phase control ( $R^2 = 0.99997$ ), high contrast ( $C > 0.9$ ), and second-order spatial convergence at  $\Delta x = 2 \text{ nm}$ . By extending the model to include Poisson–Thomas–Fermi electrostatics and a stochastic dephasing budget, we have provided quantitative design rules: the device operates with visibility  $> 0.5$  up to  $T \approx 10 \text{ K}$  for  $\mu > 10^6 \text{ cm}^2/\text{Vs}$ , and scales to a reversible XOR gate using dual independent lenses.

Immediate next steps are (i) experimental verification of the voltage-to-phase mapping in a high-mobility InGaAs/InAlAs heterostructure, (ii) integration with single-electron transistor readout for fan-out, and (iii) exploration of in-flight gate modulation for sub-picosecond logic switching. The architecture requires no superconductors, no magnetic fields, and no sub-Kelvin refrigeration—a pragmatic route beyond CMOS for ultralow-power ballistic logic.

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Entangletronic chip: flying electron + electrostatic lens gate

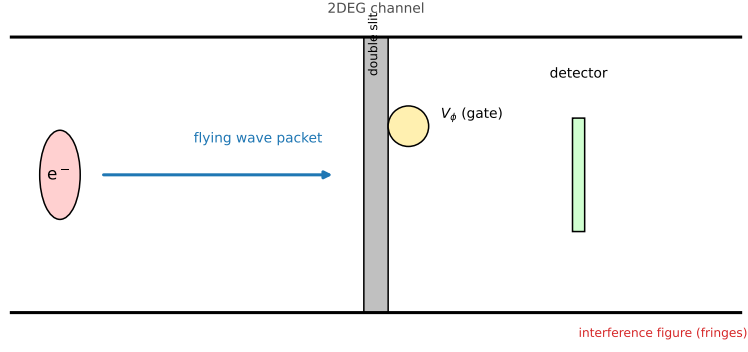


Figure 1: **EQLI device schematic.** A single electron is launched into an InGaAs 2DEG channel. A double-slit barrier splits the wave; a programmable electrostatic lens  $V_\phi$  (metallic gate finger, Sec. 2.2) behind the upper slit imparts a voltage-controlled geometric phase; a dual-bin detector reads the displaced interference figure. Inset: cross-section of the heterostructure (not to scale).

Poisson-Thomas-Fermi electrostatics: gate voltage to effective lens potential

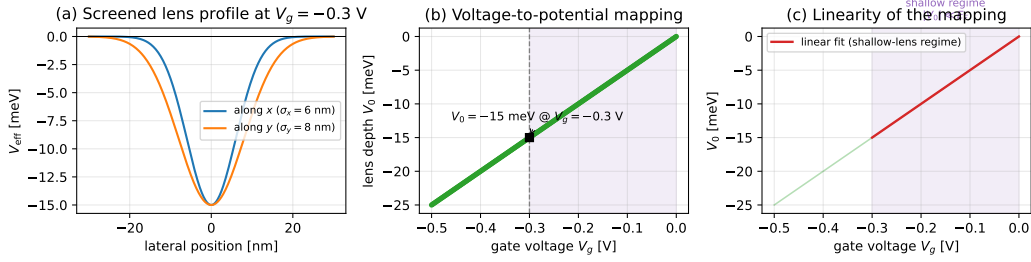


Figure 2: **Gate-voltage to effective-potential mapping.** (a) Screened effective lens profile  $V_{\text{eff}}$  at the operating point  $V_g = -0.3$  V: Gaussian parametrisation of Eq. (5) with  $V_0 \approx -15$  meV,  $\sigma_x = 6$  nm and  $\sigma_y = 8$  nm. (b) Lens depth  $V_0$  vs.  $V_g$ : linear in the shallow-lens regime (shaded,  $|V_g| \leq 0.3$  V) with a 50 meV/V coupling. (c) Linear fit in the shallow-lens regime used in the quantum simulations.

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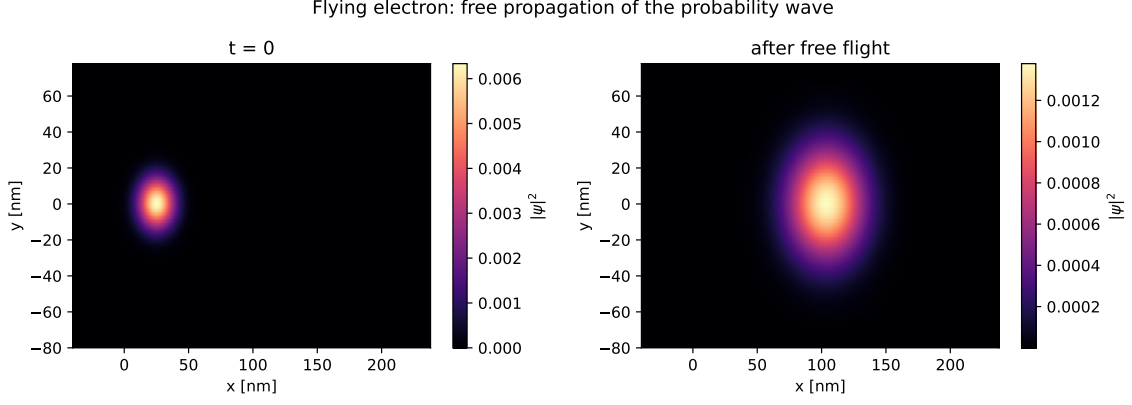


Figure 3: **Free flight benchmark.**  $|\psi|^2$  at  $t = 0$  (left) and after propagation (right): a delocalised probability wave spreading ballistically. The entrance/exit caps (not shown) are a weak perturbation ( $V_{\max}/E = 0.41$ ).

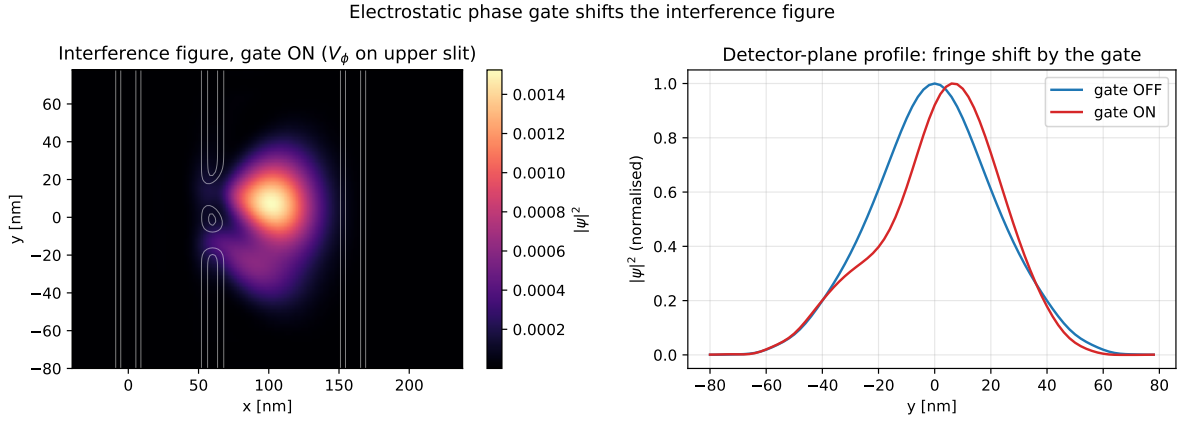


Figure 4: **Phase-gate operation.** Left: final  $|\psi|^2$  with the lens activated ( $V_g = -0.3$  V). Right: detector-plane profiles at  $x_d = 110$  nm for  $V_g = 0$  (blue) and  $V_g = -0.3$  V (red). The pure fringe shift with preserved envelope confirms geometric-phase control.

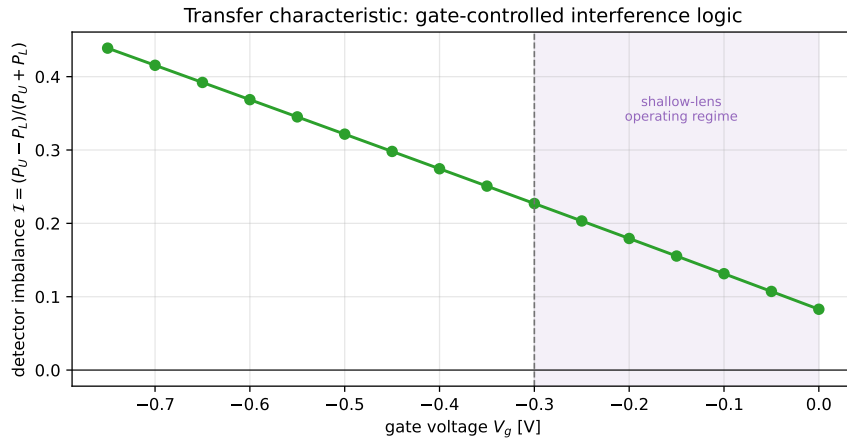


Figure 5: **Transfer characteristic.** Detector imbalance  $\mathcal{I}$  vs. gate voltage  $V_g$  (mapping  $V_g \rightarrow V_0$  of Fig. 2). The shallow-lens operating window ( $|V_g| \leq 0.3$  V, shaded) is linear,  $\mathcal{I} \approx \mathcal{I}_0 + S_V V_g$ , with  $S_V \approx -0.48$  V $^{-1}$  and  $R^2 = 0.99997$ ; the deeper-lens tail ( $|V_g| > 0.3$  V) is an extrapolation of the phase model beyond the linear-screening window.



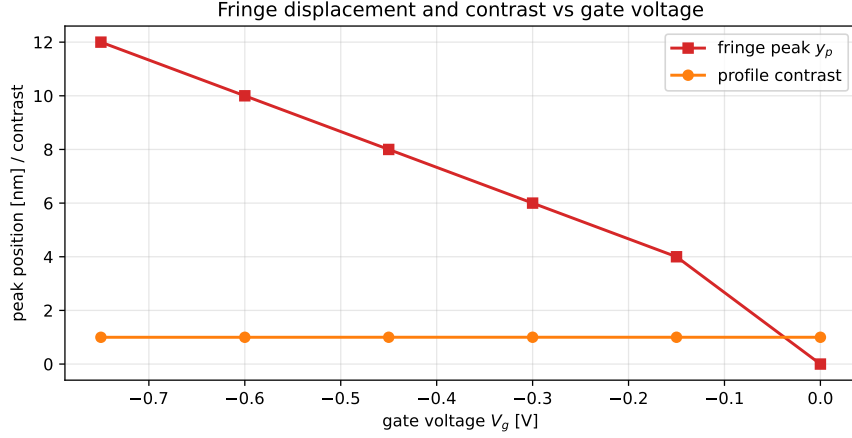


Figure 6: **Fringe displacement and contrast.** Central-peak position  $y_p$  (red, left axis) and profile contrast  $C$  (orange, right axis) vs.  $V_g$ . Contrast remains  $C > 0.9$  in the coherent limit, confirming phase-displacement (not amplitude-modulation) logic.

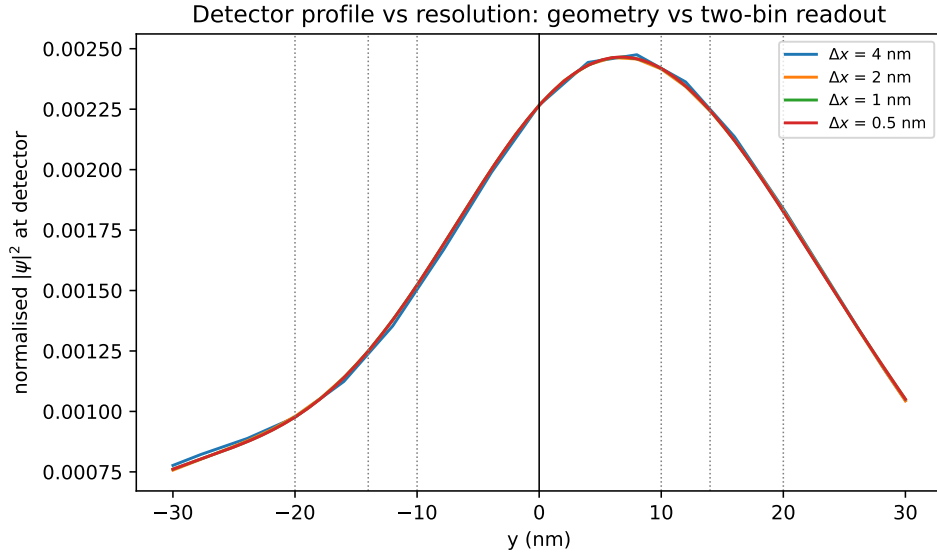


Figure 7: **Grid-sensitivity analysis.** Detector profiles  $P(y) = |\psi(x_d, y)|^2$  for  $\Delta x = \{4, 2, 1, 0.5\}$  nm. The wave-dynamics geometry (peak, centroid, width) is converged at  $\Delta x = 2$  nm. The  $L^2$  error decays  $\propto (\Delta x)^2$  (inset). The two-bin imbalance read as a continuous functional is stable ( $\approx 0.154$ ); read as a raw box-sum it drifts ( $0.31 \rightarrow 0.17$ ), proving that the apparent non-convergence is a read-out artefact, not a solver failure.

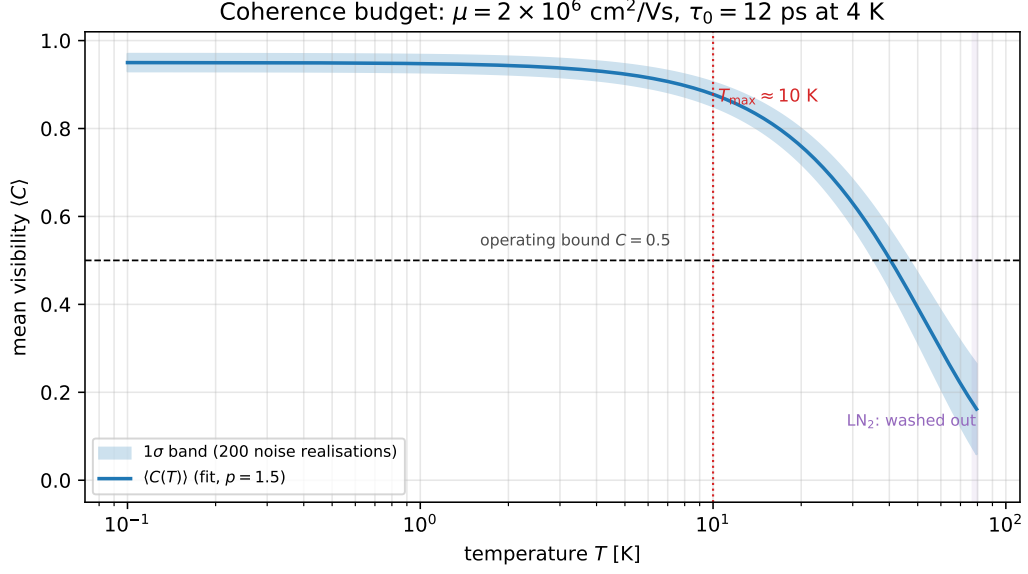


Figure 8: **Coherence budget.** Mean interference visibility  $\langle C(T) \rangle$  vs. temperature for mobility  $\mu = 2 \times 10^6 \text{ cm}^2/\text{Vs}$ . Shaded region: one standard deviation over 200 noise realisations. The dashed line is the fit of Eq. (12) with  $p = 1.5$ . The device operates with  $C > 0.5$  up to  $T \approx 10 \text{ K}$ .

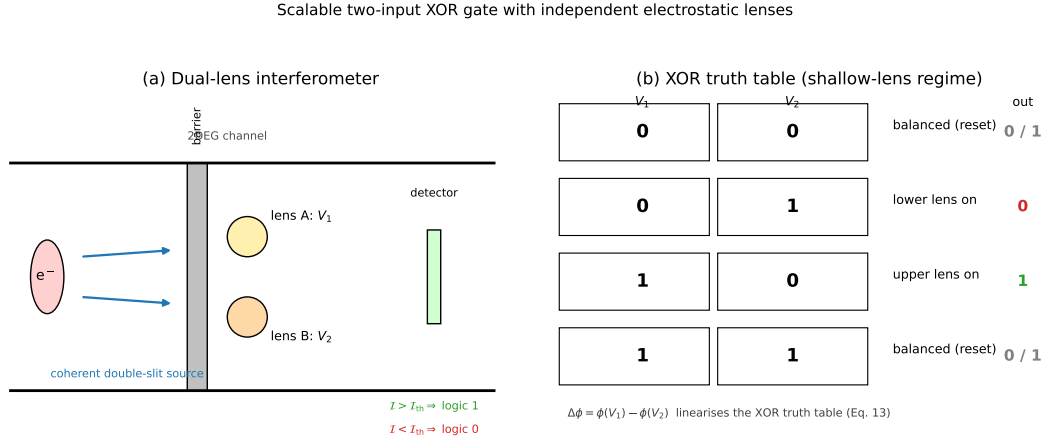


Figure 9: **Scalable XOR gate proposal.** (a) Two independent phase lenses (A and B) behind the upper and lower slits, controlled by inputs  $V_1$  and  $V_2$ . (b) Truth table: the imbalance  $\mathcal{I}$  encodes the relative phase  $\phi(V_1) - \phi(V_2)$ , realising a linearised XOR in the shallow-lens regime. (c) Layout of a  $2 \times 2$  EQLI lattice for fan-out-free reversible logic.